SEMINAR 2025

SOLUTIONS

1(a i) $\cos 36^\circ - \cos 72^\circ$

$-2\sin\left(\frac{36^\circ+72^\circ}{2}\right)\sin\left(\frac{36^\circ-72^\circ}{2}\right)$

$-2\sin 54^\circ \sin(-18^\circ)$

$\frac{2\sin 54^\circ \sin 18^\circ \cos 18^\circ}{\cos 18^\circ}$

$\frac{\sin 54^\circ \sin 36^\circ}{\cos 18^\circ}$

$\frac{-1}{2}\left[\frac{-2\sin 54^\circ \sin 36^\circ}{\cos 18^\circ}\right]$

$\frac{-1}{2}\left[\frac{\cos 90^\circ - \cos 18^\circ}{\cos 18^\circ}\right]$

$\frac{-1}{2}\left[\frac{-\cos 18^\circ}{\cos 18^\circ}\right]$

$\frac{1}{2}$.

1a(ii). Let $P = \tan^{-1}\frac{1}{4} + \tan^{-1}\frac{3}{5}$

Let $A = \tan^{-1}\frac{1}{4}$, $\tan A = \frac{1}{4}$

$B = \tan^{-1}\frac{3}{5}$, $\tan B = \frac{3}{5}$

$P = A + B$

$\tan P = \tan(A + B)$

$= \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$\tan P = \frac{\frac{1}{4} + \frac{3}{5}}{1 - (\frac{1}{4})(\frac{3}{5})}$

$\tan P = 1$

$P = \tan^{-1}(1)$

$P = \pi/4$

$\tan^{-1}\frac{1}{4} + \tan^{-1}\frac{3}{5} = \pi/4$

1b i) $2\cos^2\theta + 5\sin\theta = 1$

$2(1 - \sin^2\theta) + 5\sin\theta - 1 = 0$

$2\sin^2\theta - 5\sin\theta - 1 = 0$

$\sin\theta = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(2)(-1)}}{2(2)}$

$\sin\theta = -\frac{1}{2}, 1$

For $\sin\theta = -\frac{1}{2}$

$\theta = \sin^{-1}(-\frac{1}{2})$

$\theta = 210°, 330°$

For $\sin\theta = 1$

$\theta = \sin^{-1}(1)$

$\theta = 90°$

$\theta = \{90°, 210°, 330°\}$

1.(e) $4\cos\theta - 3\sin\theta = R\cos(\theta+\alpha)$

$4\cos\theta - 3\sin\theta = R\cos\theta\cos\alpha - R\sin\theta\sin\alpha$

$R\sin\alpha = 3$

$R\cos\alpha = 4$

$\tan\alpha = 3/4$

$\alpha = \tan^{-1}(3/4)$

$\alpha = 36.87°$

$R^2(\cos^2\alpha + \sin^2\alpha) = 3^2 + 4^2$

$R = 5$

$4\cos\theta - 3\sin\theta = 5\cos(\theta + 36.87°)$

$f(x) = 9 + 4\cos\theta - 3\sin\theta$

$= 9 + 5\cos(\theta + 36.87°)$

$f(x)_{Min} = 9 + 5(-1)$

$= 4$

$f(x)_{Min}$ occurs when $\cos(\theta + 36.87°) = -1$

$\theta + 36.87 = \cos^{-1}(-1)$

$\theta + 36.87 = 180°$

$\theta = 143.13$

$f(x) = \frac{1}{7 + 4\cos\theta - 3\sin\theta}$

$= \frac{1}{7 + 5\cos(\theta + 36.87°)}$

$= \frac{1}{7 + 5(1)}$

$= \frac{1}{12}$, it occurs when

$\cos(\theta + 36.87) = 1$

$\theta + 36.87 = 360$

$\theta = 323.13°$

16(ii) $\sin\theta + \sin 2\theta + \sin 3\theta = 0$

$\sin 3\theta + \sin\theta + \sin 2\theta = 0$

$2\sin 2\theta\cos\theta + \sin 2\theta = 0$

$\sin 2\theta\,[2\cos\theta + 1] = 0$

Either $\sin 2\theta = 0$ or $\cos\theta = -1/2$

For $\sin 2\theta = 0$

$2\theta = \sin^{-1}(0)$

$2\theta = 0°, 180°, 360°, 540°, 720°$

$\theta = 0°, 90°, 180°, 270°, 360°$

For $\cos\theta = -1/2$

$\theta = \cos^{-1}(-1/2)$

$\theta = 120°, 240°$.

(c)

2 a) $\sin\theta + 2\sin(\theta+120^\circ) + \sin(\theta+240^\circ) = 0$

$\sin\theta + \sin(\theta+240^\circ) + 2\sin(\theta+120^\circ) = 0$

$2\sin(\theta+120)\cos(-120) + 2\sin(\theta+120^\circ) = 0$

$\sin(\theta+120^\circ)\left[2\cos 120^\circ + 2\right] = 0$

$\sin(\theta+120^\circ) = 0$

$\theta + 120 = \sin^{-1}(0)$

$\theta + 120^\circ = 0^\circ, 180^\circ, 360^\circ$

$\theta = 60^\circ, 240^\circ.$

a ii) $\cos 2\theta + \cos^2\theta = 1$

$\cos 2\theta + 2\cos^2\theta - 1 = 0$

$2\cos^2 2\theta + \cos 2\theta - 2 = 0$

$$\cos 2\theta = \frac{-1 \pm \sqrt{(1)^2 - 4(2)(-2)}}{2(2)}$$

$\cos 2\theta = -1.2808, 0.7808.$

For $\cos 2\theta = -1.2808$

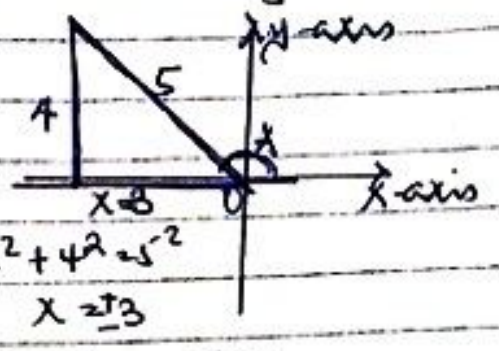
No value exists

For $\cos 2\theta = 0.7808$

$2\theta = 38.67^\circ, 321.33^\circ, 398.67^\circ, 681.33$

$\theta = 19.34^\circ, 160.67^\circ, 199.84^\circ, 340.67^\circ$

b) $\sin A = 4/5$, $\cos B = 3/5$

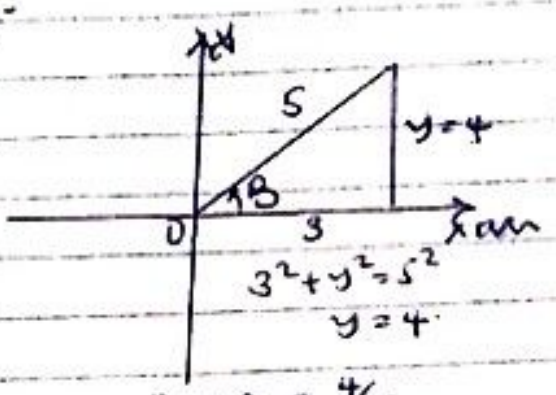


$x^2 + 4^2 = 5^2$

$x = \pm 3$

$\cos A = -3/5$

$\tan A = -4/3$



$3^2 + y^2 = 5^2$

$y = 4$

$\sin B = 4/5$

$\tan B = 4/3$

i) $\sin(A - B) = \sin A \cos B - \cos A \sin B$

$= 4/5 \cdot 3/5 - (-3/5)(4/5)$

$= \frac{24}{25}$

ii) $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$= \frac{(-4/3) - (4/3)}{1 - (-4/3)(4/3)}$

$= \frac{24}{25}$

c) $\sin A + \sin B + \sin C$

$2\sin\frac{A+B}{2}\cos\frac{A-B}{2} + 2\sin\frac{C}{2}\cos\frac{C}{2}$

$\frac{A+B}{2} = 90 - \frac{C}{2}$.

$\sin\left(\frac{A+B}{2}\right) = \sin\left(90 - \frac{C}{2}\right) = \cos\frac{C}{2}$.

$\cos\left(\frac{A+B}{2}\right) = \cos\left(90 - \frac{C}{2}\right) = \sin\frac{C}{2}$.

$2\cos\frac{C}{2}\cos\frac{A-B}{2} + 2\sin\frac{C}{2}\cos\frac{C}{2}$

$2\cos\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right) + \sin\frac{C}{2}\right]$

$2\cos\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right) + \cos\frac{A+B}{2}\right]$.

$2\cos\frac{C}{2}\left[2\cos\frac{A}{2}\cos\frac{B}{2}\right]$.

$4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$.

$\cos A + \cos B + \cos C$

$2\cos\frac{A+B}{2}\cos\frac{A-B}{2} + \cos C$

$2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) + 2\cos^2\frac{C}{2} - 1$

$\frac{A+B}{2} = 90 - \frac{C}{2}$

$\cos\left(\frac{A+B}{2}\right) = \cos\left(90 - \frac{C}{2}\right) = \sin\frac{C}{2}$

$\sin\left(\frac{A+B}{2}\right) = \sin\left(90 - \frac{C}{2}\right) = \cos\frac{C}{2}$

$2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) + 1 - 2\sin^2\frac{C}{2}$

$2\sin\frac{C}{2}\cos\left(\frac{A-B}{2}\right) - 2\sin^2\frac{C}{2} + 1$

$2\sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right) - \sin\frac{C}{2}\right] + 1$

$2\sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right) - \cos\left(\frac{A+B}{2}\right)\right] + 1$

$1 + 2\sin\frac{C}{2}\left[-2\sin\frac{A}{2}\sin\left(\frac{-B}{2}\right)\right]$

$1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}.$

3(a) $2\cos(\theta+30^\circ) = 3\sin(\theta-30^\circ)$

$2[\cos\theta\cos 30^\circ - \sin\theta\sin 30^\circ] = 3(\sin\theta\cos 30 - \cos\theta\sin 30)$

$2\cos\theta\cos 30^\circ - 2\sin\theta\sin 30^\circ = 3\sin\theta\cos 30^\circ - 3\cos\theta\sin 30^\circ$

$\cos\theta[2\cos 30^\circ + 3\sin 30^\circ] = \sin\theta[3\cos 30^\circ + 2\sin 30^\circ]$

$$\frac{\sin\theta}{\cos\theta} = \frac{2\cos 30^\circ + 3\sin 30^\circ}{3\cos 30^\circ + 2\sin 30^\circ}$$

$$\tan\theta = \frac{2+3\tan 30^\circ}{3+2\tan 30}$$

$\tan\theta = 0.8983^\circ$

$\theta = \tan^{-1}(0.8983)$

$\theta = 41.93^\circ$.

b) $4\tan\theta + 3\sec\theta = \cos\theta$

$$\frac{4\sin\theta}{\cos\theta} + \frac{3}{\cos\theta} = \cos\theta$$

$4\sin\theta + 3 = \cos^2\theta$

$4\sin\theta + 3 = 1 - \sin^2\theta$

$\sin^2\theta + 4\sin\theta + 2 = 0$

$$\sin\theta = \frac{-4 \pm \sqrt{4^2 - 4(1)(2)}}{2(1)}$$

$\sin\theta = -3.4142, -0.5858$

For $\sin\theta = -3.4142$ Discard

no value of θ exist.

For $\sin\theta = -0.5858$

$\theta = \sin^{-1}(-0.5858)$

$\theta = 215.86^\circ, 324.14^\circ$

3.b(i) $x = \cos\theta, \quad y = 3\sin 2\theta$

$y = 3\sin 2\theta$

$y = 3(2\sin\theta\cos\theta)$

$y = 6\cos\theta\sqrt{1-\cos^2\theta}$

$y = 6x\sqrt{1-x^2}$

ii) $x = \tan\theta + \cos\theta \quad --(i)$

$y = \tan\theta - \cos\theta \quad --(ii)$

(i)+(ii) $\Rightarrow x + y = 2\tan\theta$

(i)-(ii) $\Rightarrow x - y = 2\cos\theta$

$\tan\theta = \frac{x+y}{2}, \quad \cos\theta = \frac{x-y}{2}$

$\sec\theta = \frac{2}{x-y}$

$1 + \tan^2\theta = \sec^2\theta$

$1 + \left(\frac{x+y}{2}\right)^2 = \left(\frac{2}{x-y}\right)^2.$

$\frac{4}{(x-y)^2} - \frac{(x+y)^2}{4} = 1.$

3 c) $\frac{\cos\theta - 1}{\sec\theta + \tan\theta} + \frac{\cos\theta + 1}{\sec\theta - \tan\theta}$

$\frac{(\cos\theta - 1)(\sec\theta - \tan\theta) + (\cos\theta + 1)(\sec\theta + \tan\theta)}{(\sec\theta + \tan\theta)(\sec\theta - \tan\theta)}$

$\frac{(1 - \sin\theta - \sec\theta + \tan\theta) + (1 + \sin\theta + \sec\theta + \tan\theta)}{(\sec^2\theta - \tan^2\theta)}$.

$\frac{2(1 + \tan\theta)}{\sec^2\theta - \tan^2\theta}$.

But $\sec^2\theta - \tan^2\theta = 1$

$\Rightarrow 2(1 + \tan\theta)$.

3 (ii)

$\tan^2\theta - \sin^2\theta = \sin^4\theta \sec^2\theta$

$\frac{\sin^2\theta}{\cos^2\theta} - \sin^2\theta$

$\sin^2\theta \left(\frac{1}{\cos^2\theta} - 1\right)$.

$\sin^2\theta \left(\frac{1 - \cos^2\theta}{\cos^2\theta}\right)$

$\sin^2\theta \cdot \frac{\sin^2\theta}{\cos^2\theta}$.

$\sin^4\theta \sec^2\theta$.

No. 6(a) Let $y = \sin 2x$

$y + \delta y = \sin 2(x+\delta x)$

$\delta y = \sin 2(x+\delta x) - \sin 2x$

$\frac{\delta y}{\delta x} = \frac{2\cos(2x+\delta x)\sin\delta x}{\delta x}$

As $\lim_{\delta x \to 0} \frac{\delta y}{\delta x} \simeq \frac{dy}{dx}$, $\sin\delta x = \delta x$

$\frac{dy}{dx} = 2\cos 2x.$

Let $y = e^{2x}$

$y + \delta y = e^{2(x+\delta x)}$

$\delta y = e^{2(x+\delta x)} - e^{2x}$

$= e^{2x}\left[e^{2\delta x} - 1\right]$

But $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

$\delta y = e^{2x}\left\{\left[1 + (2\delta x) + \frac{(2\delta x)^2}{2!} + \cdots\right] - 1\right\}$

$\frac{\delta y}{\delta x} = e^{2x}\left\{\left[1 + 2\delta x + 2(\delta x)^2 + \cdots\right] - 1\right\}$

δx

$= e^{2x}\left[2 + (\delta x)^2 + \cdots\right]$

As $\lim_{\delta x \to 0} \frac{\delta y}{\delta x} \simeq \frac{dy}{dx}$.

$\frac{dy}{dx} = 2e^{2x}.$

b) i) $\frac{d}{dx}(x^2+6x+5) = 2x+6$

ii) e^{2x+3}

Let $y = e^{2x+3}$, $u = 2x+3$; $y = e^u$

$\frac{du}{dx} = 2$, $\frac{dy}{du} = e^u$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$= e^u \cdot 2$

$= 2e^{2x+3}$.

iii) $y = \sqrt{\cos 2x}$, $t = \cos 2x$ $y = t^{1/2}$

$\frac{dt}{dx} = -2\sin 2x$, $\frac{dy}{dt} = \frac{1}{2}t^{-1/2}$

$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$

$= \frac{1}{2}t^{-1/2} \cdot -2\sin 2x$

$\frac{d}{dx}(\sqrt{\cos 2x}) = \frac{-\sin 2x}{\sqrt{\cos 2x}}$

6e) Let $y = \sqrt{x}$

$y + \delta y = \sqrt{x + \delta x}$

$x = 9, \ \delta x = 0.8$

$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$

$\delta y \simeq \frac{dy}{dx} \cdot \delta x$

$\simeq \frac{1}{2\sqrt{9}} \cdot (0.8)$

$\simeq \frac{4}{30}$

$\sqrt{9.88} \simeq y + \delta y$

But $y = \sqrt{9} = 3$

$\sqrt{9.8} \approx 3\frac{4}{30}$

$\approx 3\frac{2}{15}$

w) $y = \ln \theta$

$y + \delta y = \ln(\theta + \delta\theta)$

$\frac{dy}{d\theta} = \cos\theta$

$\theta = 60^\circ \quad \delta\theta = -0.5^\circ = -\frac{0.5\pi}{180}$

$=$

$\delta y \simeq \frac{dy}{d\theta} \cdot \delta\theta$

$\simeq (\cos 60^\circ) \frac{-0.5\pi}{180}$

$\simeq -0.00436$

$\sin 59.5^\circ = y + \delta y$

$= \sin 60^\circ - 0.00436$

$= 0.86603 - 0.00436$

$\simeq 0.86167$

$\simeq 0.862$ (3dps)

$f(x) = x^2 - 4x + 3$

$f'(x) = 2x - 4$

At turning point $f'(x) = 0$

$2x - 4 = 0$

$x = 2$

$f''(x) = 2 > 0$

Minimum value occurs

when $x = 2$, $f(2) = 2^2 - 4(2) + 3 = -1$

$(2, -1)$ is Minimum turning point

when $f(x) = 0$, $x^2 - 4x + 3 = 0$

$(x-3)(x-1) = 0$

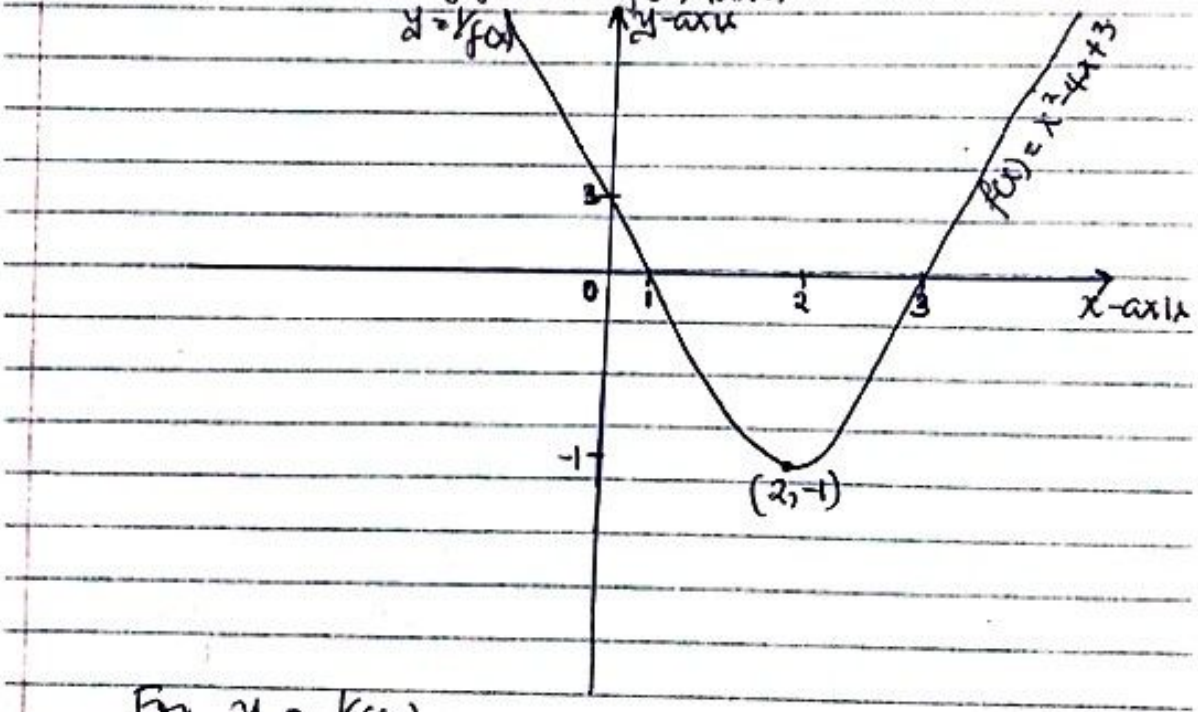
$x = 1, x = 3$

$(1, 0), (3, 0)$

when $x = 0$, $f(0) = 0^2 - 4(0) + 3 = 3$

$(0, 3)$

Sketch of $f(x)$ and $f(x)$-axis



For $y = 1/f(x)$

$(2, -1)$ is a Maximum turning point.

Intercepts $(0, 1/3)$

	$x < 1$	$1 < x < 3$	$x > 3$
$f(x)$	+	−	+
$y = 1/f(x)$	+	−	+

7(c) $f(x) = (1-x)^{-1/2}$ $\quad f(0) = 1$

$f'(x) = -\frac{1}{2}(1-x)^{-3/2} \cdot -1$

$= \frac{1}{2}(1-x)^{-3/2}$ $\quad f'(0) = \frac{1}{2}$

$f''(x) = -\frac{3}{4}(1-x)^{-5/2} \cdot (-1)$ $\quad f''(0) = \frac{3}{4}$

$= \frac{3}{4}(1-x)^{-5/2}$

$$\frac{1}{\sqrt{1-x}} = 1 + \frac{1}{2}x + \frac{3}{4} \cdot \frac{1}{2!}x^2,$$

$$= 1 + \frac{x}{2} + \frac{3x^2}{8},$$

When $x = -2$, $y = 6 - 3(-2) = 12$ $(-2, 12)$

When $x = 2$, $y = 6 - 3(2) = 0$ $(2, 0)$

T6(i) $A = \int_{-2}^{2} y_1 - y_2 \, dx$

$= \int_{-2}^{2} (6 - 3x) - (x^2 - 3x + 2) \, dx$

$= \int_{-2}^{2} 4 - x^2 \, dx$

$= \left[4x - \frac{x^3}{3} \right]_{-2}^{2}$

$= \left(4(2) - \frac{2^3}{3} \right) - \left(4(-2) - \frac{(-2)^3}{3} \right)$

$A = \frac{32}{3}$ Sq Units

Area bounded by the curve $y = x^2 - 3x + 2$ and the line $3x + y = 6$ is $\frac{32}{3}$ Sq Units.

T6(ii) $V = \pi \int_{-2}^{2} y_1^2 - y_2^2 \, dx$

$= \pi \int_{-2}^{2} (6 - 3x)^2 - (x^2 - 3x + 2)^2 \, dx$

$= \pi \int_{-2}^{2} 32 - 24x - 4x^2 + 6x^3 - x^4 \, dx$

$V = \pi \left[32x - 12x^2 - \frac{4}{3}x^3 + \frac{6}{4}x^4 - \frac{x^5}{5} \right]_{-2}^{2}$

$V = \pi \left\{ \left(64 - 48 - \frac{32}{3} - 24 - \frac{32}{5} \right) - \left(-64 - 48 - \frac{32}{3} - 24 + \frac{32}{5} \right) \right\}$

$V = \frac{1408\pi}{15}$ Cubic Units

iii) $y = e^{x^2+2x}$

$\ln y = \ln e^{x^2+2x}$

$\ln y = x^2+2x$

$\frac{1}{y}\frac{dy}{dx} = 2x+2$

$\frac{dy}{dx} = (2x+2)\, e^{x^2+2x}$.

(iv) $y = x^2 + x^{\sin x}$.

$u = x^2$ $\qquad v = x^{\sin x}$

$\frac{du}{dx} = 2x$ $\qquad \ln v = \ln x^{\sin x}$

$\ln v = (\sin x)\ln x$

$\frac{1}{v}\frac{dv}{dx} = \sin x \cdot \frac{1}{x} + (\ln x)\cos x$

$\frac{dv}{dx} = \frac{\sin x}{x} + (\ln x)(\cos x)\, x^{\sin x}$

$\frac{dy}{dx} = 2x + \left[\frac{\sin x}{x} + \ln x)\cos x\right] x^{\sin x}$.

No. 8

i) Let $y = \dfrac{e^{2x}\sin x}{(1+x^2)(1-x)}$.

$$\ln y = \ln\left[\frac{e^{2x}\sin x}{(1+x^2)(1-x)}\right]$$

$$\ln y = \ln e^{2x} + \ln\sin x - \ln(1+x^2) - \ln(1-x)$$

$$\ln y = 2x + \ln\sin x - \ln(1+x^2) - \ln(1-x)$$

$$\frac{1}{y}\frac{dy}{dx} = 2 + \frac{\cos x}{\sin x} - \frac{2x}{1+x^2} - \frac{(-1)}{1-x}$$

$$\frac{dy}{dx} = \left(2 + \cot x - \frac{2x}{1+x^2} + \frac{1}{1-x}\right) y$$

$$\frac{dy}{dx} = \left(2 + \cot x - \frac{2x}{1+x^2} + \frac{1}{1-x}\right)\left(\frac{e^{2x}\sin x}{(1+x^2)(1-x)}\right)$$

ii) Let $y = x^3 \log_{10}(1+x^2)$

$u = x^3$ $\qquad$ $v = \log_{10}(1+x^2)$

$\dfrac{du}{dx} = 3x^2$

$10^v = 1+x^2$

$\ln 10^v = \ln(1+x^2)$

$v \ln 10 = \ln(1+x^2)$

$$(\ln 10)\frac{dv}{dx} = \frac{2x}{1+x^2}$$

$$\frac{dv}{dx} = \frac{2x}{(1+x^2)\ln 10}.$$

$$\frac{dy}{dx} = x^3 \cdot \frac{2x}{(1+x^2)\ln 10} + 3x^2 \log_{10}(1+x^2)$$

8(b) Equation of tangent & normal

$$y = x^2 - 3\log_2 x + 5$$

$$y = x^2 - 3\frac{\log_e x}{\log_e 2} + 5$$

$$\frac{dy}{dx} = 2x - \frac{3}{x \ln 2}$$

$$\left.\frac{dy}{dx}\right|_{(2,6)} = 2(2) - \frac{3}{2\ln 2}$$

$$\approx 1.836$$

$$\frac{y-6}{x-2} = 1.836$$

$$y - 6 = 1.836x - 3.672$$

$$y = 1.836x + 2.328.$$

(ii) $$y = \frac{x-2}{x+1}$$

$$\frac{dy}{dx} = \frac{(x+1)(1) - (x-2)(1)}{(x+1)^2}$$

$$= \frac{3}{(x+1)^2}$$

$$\left.\frac{dy}{dx}\right|_{(-2,4)} = \frac{3}{(-2+1)^2} = 3$$

Gradient of tangent = 3

Gradient of the normal = $-\frac{1}{3}$

Equation of tangent

$$\frac{y-4}{x+2} = 3$$

$$y = 3x + 10.$$

Equation of the normal

$$\frac{y-4}{x+2} = \frac{-1}{3}$$

$$x + 3y = 10.$$

8c) $\delta r = \frac{6}{100} r.$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dv}{dr} = 4\pi r^2.$$

$$\delta v = \frac{dv}{dr} \cdot \delta r.$$

$$= 4\pi r^2 \cdot \frac{6r}{100}.$$

$$\% \delta v = \frac{\delta v}{v} \times 100\%$$

$$= \left(\frac{\frac{24\pi r^3}{100}}{\frac{4}{3}\pi r^3} \right) \times 100\%$$

$$= 18\%$$

8(d)



$2x + y = 2000$

$y = 2000 - 2x$

$A = xy$

$A = x(2000 - 2x)$

$A = 2000x - 2x^2$

$\frac{dA}{dx} = 2000 - 4x$

If Max/Min $\frac{dA}{dx} = 0$

$2000 - 4x = 0$

$x = 500$

$\frac{d^2A}{dx^2} = -4 < 0$

Maximum Area exists.

When $x = 500$, $y = 2000 - 2(500)$

$y = 1000$

$A_{max} = (500)(1000)$

$= 500{,}000\ m^2$.

Maximum possible Area is $500{,}000\ m^2$

No-9 $\int_0^{\pi/2} x \sin x \, dx$

Sign	Differentiation	Integration
+	x	$\sin x$
−	1	$-\cos x$
+	0	$-\sin x$

$$\int_0^{\pi/2} x \sin x \, dx = \left[-x \cos x + \sin x \right]_0^{\pi/2}$$

$$= \left(\frac{-\pi}{2} \cos\frac{\pi}{2} + \sin\frac{\pi}{2} \right) - (0)$$

$$= 1$$

Alternatively

let $u = x$ $\quad \frac{dv}{dx} = \sin x$

$\frac{du}{dx} = 1$ $\quad v = -\cos x$

$$\int_0^{\pi/2} x \sin x \, dx = \left[-x \cos x \right]_0^{\pi/2} + \int_0^{\pi/2} \cos x \, dx$$

$$= \left[-x \cos x + \sin x \right]_0^{\pi/2}$$

$$= (1) - (0)$$

$$= 1$$

9(ii) $\int_0^2 \frac{dx}{4+x^2}$

Let $4+x^2 = 4+4\tan^2 u = 4\sec^2 u$

$x = 2\tan u$

$dx = 2\sec^2 u\, du$

x	0	2
u	0	$\pi/4$

$$\int_0^{\pi/4} \frac{2\sec^2 u\, du}{4\sec^2 u}$$

$$\frac{1}{2}\int_0^{\pi/4} du$$

$$\left[\frac{u}{2}\right]_0^{\pi/4}$$

$$\left(\frac{\pi}{8}\right) - \left(0\right)$$

$$\frac{\pi}{8}$$

9 (a) (ii)

$$\int_0^{\ln 2} \frac{e^x dx}{1+e^x}$$

$u = 1 + e^x$

$du = e^x dx$

x	0	$\ln 2$
u	2	3

$$\int_2^3 \frac{du}{u}$$

$$[\ln u]_2^3$$

$$\ln 3 - \ln 2$$

$$\ln(3/2)$$

(b) $\frac{dy}{dx} + y = e^{2x}$.

$$I = e^{\int dx}$$

$$I = e^x$$

$$e^x \frac{dy}{dx} + e^x y = e^{3x}.$$

$$\frac{d}{dx}(e^x y) = e^{3x}$$

$$\int \frac{d}{dx}(e^x y)\,dx = \int e^{3x} dx$$

$$e^x y = \frac{e^{3x}}{3} + c$$

9 b(ii)

$$\frac{dy}{dx} = \frac{x \ln x}{y}$$

$$\int y\, dy = \int x \ln x\, dx$$

$$\frac{y^2}{2} = \int x \ln x\, dx$$

Sign	Differentiation	Integration
+	$\ln x$	x
−	$1/x$	$x^2/2$

$$\frac{y^2}{2} = \frac{x^2}{2} \ln x - \int \frac{1}{x} \cdot \frac{x^2}{2}\, dx$$

$$= \frac{x^2}{2} \ln x - \int \frac{x}{2}\, dx$$

$$\frac{y^2}{2} = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$$

9 b (iii)

$$x\frac{dy}{dx} + y = \cos x$$

$$\frac{d}{dx}(xy) = \cos x$$

$$\int \frac{d}{dx}(xy)\,dx = \int \cos x\,dx$$

$$xy = \sin x + c$$

9 (c) $\cdot$ $\quad \frac{dy}{dx} = x + y$

$$\frac{dy}{dx} - y = x.$$

$$I = e^{\int -dx}$$

$$= e^{-x}$$

$$e^{-x}\frac{dy}{dx} - e^{-x}y = xe^{-x}$$

$$\frac{d}{dx}\left(e^{-x}y\right) = xe^{-x}$$

$$\int \frac{d}{dx}(e^{-x}y)\,dx = \int xe^{-x}\,dx$$

$$e^{-x}y = \int xe^{-x}\,dx.$$

Sign	Differentiate	Integrate
+	x	e^{-x}
−	1	$-e^{-x}$
+	0	e^{-x}

$$e^{-x}y = -xe^{-x} - e^{-x} + c.$$

$$y = -x - 1 + ce^{x}.$$

$x = 0$, $y = 3$

$$3 = -1 + c$$

$$c = 4.$$

9 b (iii)

$$x\frac{dy}{dx} + y = \cos x$$

$$\frac{d}{dx}(xy) = \cos x$$

$$\int \frac{d}{dx}(xy)\,dx = \int \cos x\,dx$$

$$xy = \sin x + c$$

9 (c) .

$$\frac{dy}{dx} = x + y$$

$$\frac{dy}{dx} - y = x.$$

$$I = e^{\int -dx}$$

$$= e^{-x}$$

$$e^{-x}\frac{dy}{dx} - e^{-x}y = xe^{-x}$$

$$\frac{d}{dx}(e^{-x}y) = xe^{-x}$$

$$\int \frac{d}{dx}(e^{-x}y)\,dx = \int xe^{-x}\,dx$$

$$e^{-x}y = \int xe^{-x}\,dx.$$

Sign	Differentiate	Integrate
+	x	e^{-x}
−	1	$-e^{-x}$
+	0	e^{-x}

$$e^{-x}y = -xe^{-x} - e^{-x} + c.$$

$$y = -x - 1 + ce^{x}.$$

$x = 0$, $y = 3$

$$3 = -1 + c$$

$$c = 4.$$

$$y = -x - 1 + 4e^{x}$$

When $x = \ln 2$

$$y = -\ln 2 - 1 + 4e^{\ln 2}$$

$$y = -\ln 2 - 1 + 8$$

$$y = 7 - \ln 2$$

No. 10 a(i)

Let

$$y = \log_e \left[\frac{(1+x^2)(4-x)^2}{\sqrt{1-x^2}} \right]$$

$$y = \log_e (1+x^2) + \log_e (4-x)^2 - \log_e (1-x^2)^{1/2}$$

$$y = \log_e (1+x^2) + 2\log_e (4-x) - \tfrac{1}{2}\log_e (1-x^2)$$

$$\frac{dy}{dx} = \frac{2x}{1+x^2} + \frac{2(-1)}{4-x} - \frac{1}{2}\frac{(-2x)}{(1-x^2)}.$$

$$\frac{dy}{dx} = \frac{2x}{1+x^2} - \frac{2}{4-x} + \frac{x}{1-x^2}.$$

10 (a)(ii)

Let $y = \tan^{-1}\left(\frac{1+x}{1-x}\right)$

$$\tan y = \frac{1+x}{1-x}$$

$$\sec^2 y \frac{dy}{dx} = \frac{(1-x)(1) - (1+x)(-1)}{(1-x)^2}$$

$$(1+\tan^2 y)\frac{dy}{dx} = \frac{2}{(1-x)^2}.$$

$$\left[\frac{(1+x)^2}{1-x}\right]\frac{dy}{dx} = \frac{2}{(1-x)^2}.$$

$$\frac{2x^2}{(1-x)^2}\frac{dy}{dx} = \frac{2}{(1-x)^2}$$

$$\frac{dy}{dx} = \frac{1}{1+x^2}.$$

(iii)

Let $y = \frac{x \sin x}{1+e^x}$.

$\frac{dy}{dx} = (1+e^x)$

$\ln y = \ln\left(\frac{x \sin x}{1+e^x}\right)$

$\ln y = \ln x + \ln \sin x - \ln(1+e^x)$

$\frac{1}{y}\frac{dy}{dx} = \frac{1}{x} + \frac{\cos x}{\sin x} - \frac{e^x}{1+e^x}$

$\frac{dy}{dx} = \left(\frac{1}{x} + \cot x - \frac{e^x}{1+e^x}\right)\left(\frac{x \sin x}{1+e^x}\right)$

10(b) $y = 4^{2x+1}$

$\ln y = \ln 4^{2x+1}$

$\ln y = (2x+1)\ln 4$

$\frac{1}{y}\frac{dy}{dx} = 2\ln 4$.

$\frac{dy}{dx} = (2\ln 4)\, 4^{2x+1}$

$\int 4^{2x+1}\, dx = \frac{4^{2x+1}}{2\ln 4} + C$

$x = t + 1/t$

$5/2 = t + 1/t$

$2t^2 - 5t + 2 = 0$

$t = \frac{5 \pm \sqrt{(-5)^2 - 4(2)(2)}}{2(2)}$

$t = 2, 1/2$

When $t = 2$

$y = 2 - 1/2 = 3/2$. $(2, 3/2)$

$\therefore t = 2$

$\left.\frac{dy}{dx}\right|_{t=2} = \frac{1 + (2)^2}{(2)^2 - 1} = \frac{5}{3}$.

$\frac{y - 3/2}{x - 5/2} = \frac{5}{3}$

$5x - 3y = 8$

No. 11.

(a) $y = \frac{\cos x}{x}$

$xy = \cos x$

$x\frac{dy}{dx} + y(1) = -\sin x$

$x\frac{d^2y}{dx^2} + \frac{dy}{dx} + \frac{dy}{dx} = -\cos x$

$x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + xy = 0$

(b).

$x = t^3 - 3t + 4$, $y = t^2 - 4t$

$\frac{dx}{dt} = 3t^2 - 3$, $\frac{dy}{dx} = 2t - 4$

$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$

$= 2t - 4 \cdot \frac{1}{3t^2 - 3}$

$= \frac{2t - 4}{3t^2 - 3}$

b(ii) At turning point $\frac{dy}{dx} = 0$

$\frac{2t - 4}{3t^2 - 3} = 0$

$t = 2$.

$x = (2)^3 - 3(2) + 4$, $y = 2^2 - 4(2) + 1$

$x = 6$, $y = -5$.

Turning point is $(6, -5)$

No. 11 (c)

$x = \frac{1+t}{2+t}$; $y = \frac{t}{3+t}$

$2x + xt = 1 + t$

$t(x-1) = 1 - 2x$

$t = \frac{1-2x}{x-1}$

$y = \frac{\left(\frac{1-2x}{x-1}\right)}{3 + \left(\frac{1-2x}{x-1}\right)}$

$y = \frac{1-2x}{x-2}$.

$\frac{dy}{dx} = \frac{(x-2)(-2) - (1-2x)(1)}{(x-2)^2}$

$= \frac{3}{(x-2)^2}$

At turning point $\frac{dy}{dx} = 0$

$\frac{3}{(x-2)^2} = 0$

no value of x exists for turning point.

Vertical Asymptote

$x = 2$.

Horizontal Asymptote

$y = \frac{1/x - 2}{1 - 2/x}$

As $x \to \infty$, $y = -2$.

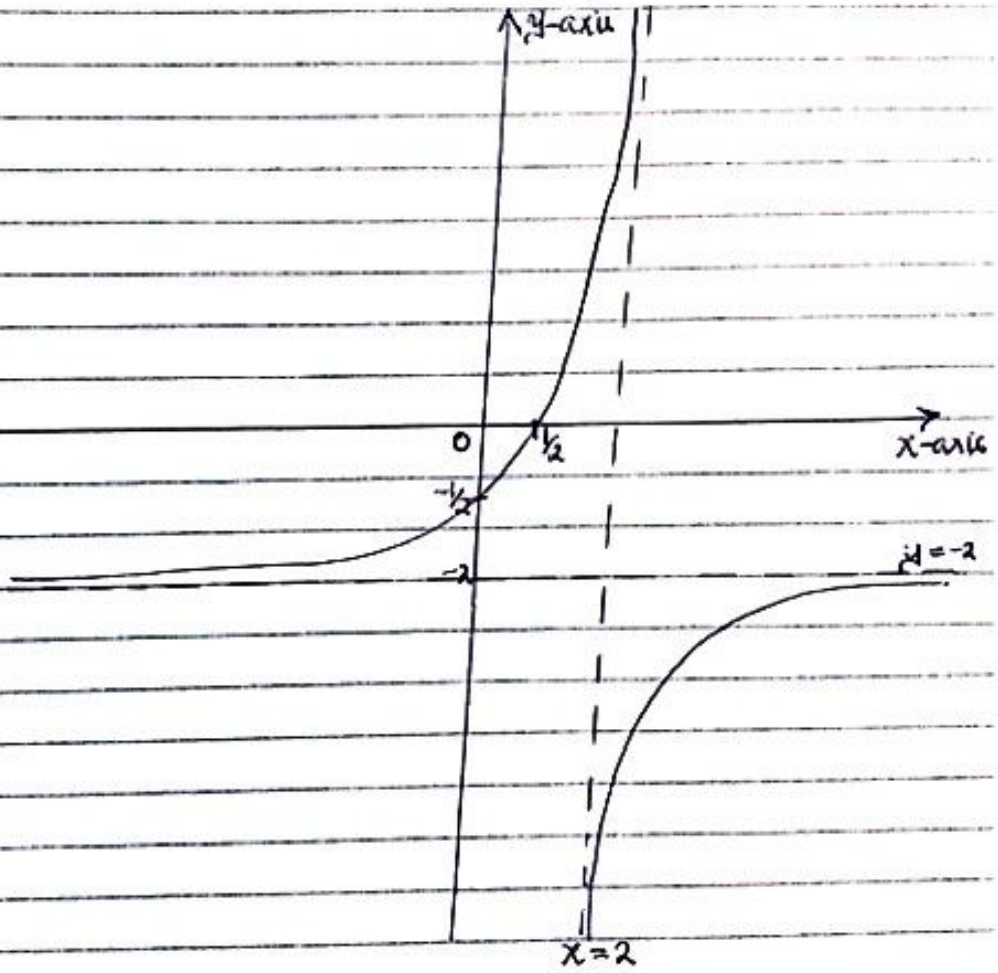
When $x = 0$, $y = -1/2$ $(0, -1/2)$

When $y = 0$, $x = 1/2$ $(1/2, 0)$

Critical values $x = \frac{1}{2}, 2$

	$x < \frac{1}{2}$	$\frac{1}{2} < x < 2$	$x > 2$
$y = \frac{1-2x}{x-2}$	−	+	−

Sketch $y = \frac{1-2x}{x-2}$



No. 12

(a) $y = \sqrt{\frac{1+\sin x}{1-\sin x}}$

$y^2 = \frac{1+\sin x}{1-\sin x}$

$2y\frac{dy}{dx} = \frac{(1-\sin x)(\cos x) - (1+\sin x)(-\cos x)}{(1-\sin x)^2}$

$= \frac{(\cos x)\left[1-\sin x + 1+\sin x\right]}{(1-\sin x)^2}$

$2y\frac{dy}{dx} = \frac{2\cos x}{(1-\sin x)^2}$

$\frac{dy}{dx} = \frac{\cos x}{(1-\sin x)^2} \cdot \left(\frac{1-\sin x}{1+\sin x}\right)^{1/2}$

$= \frac{\cos x}{(1-\sin x)^2} \cdot \left\{\frac{(1-\sin x)(1-\sin x)}{(1+\sin x)(1-\sin x)}\right\}$

$= \frac{\cos x}{(1-\sin x)^2} \cdot \frac{(1-\sin x)}{(\cos^2 x)^{1/2}}$

$= \frac{1}{1-\sin x}$

(b) $\int_0^{\sqrt{\pi/2}} x\cos(x^2)\,dx$

$u = x^2$

$du = 2x\,dx \Rightarrow \frac{du}{2} = x\,dx$

x	0	$\sqrt{\pi/2}$
u	0	$\pi/2$

$\int_0^{\pi/2} \cos u\,du$

$\left[\sin u\right]_0^{\pi/2}$

$(1) - (0)$

1

12 b(ii)

$$\int_0^{\pi/4} \cos^3 x \sin x \, dx$$

$u = \cos x$

$du = -\sin x \, dx$

x	0	$\pi/4$
u	1	$\frac{\sqrt{2}}{2}$

$$-\int_1^{\frac{\sqrt{2}}{2}} u^3 \, du$$

$$-\left[\frac{u^4}{4}\right]_1^{\frac{\sqrt{2}}{2}}$$

$$\frac{-1}{4}\left\{\left(\frac{\sqrt{2}}{2}\right)^4 - (1)^4\right\}$$

$$\frac{3}{16}$$

12 b(iii)

$$\int_0^{\frac{\pi}{2}} \frac{5 \, d\theta}{3 + 4\cos\theta}$$

12 6(iii) $\int_0^{\pi/2} \frac{5\,d\theta}{5+4\cos\theta}$

$t = \tan\theta/2$

$\cos\theta = \frac{1-t^2}{1+t^2}$

$dt = \frac{1}{2}\sec^2\theta/2\, d\theta$

$\frac{2\,dt}{1+t^2} = d\theta$

θ	0	$\pi/2$
t	0	1

$$\int_0^{\pi/2} \frac{5\,d\theta}{5+4\cos\theta} = 5\int_0^1 \frac{\frac{2dt}{1+t^2}}{5+4\left(\frac{1-t^2}{1+t^2}\right)}$$

$$= 5\int_0^1 \frac{2dt}{5(1+t^2)+4(1-t^2)}$$

$$= 10\int_0^1 \frac{dt}{9+t^2}$$

Let $t = 3\tan u$

$9+t^2 = 9+9\tan^2 u = 9\sec^2 u$

$dt = 3\sec^2 u\, du$

t	0	1
u	0	$\pi/4$

$$= 10\int_0^{\pi/4} \frac{3\sec^2 u\, du}{9\sec^2 u}$$

$$= \frac{10}{3}\int_0^{\pi/4} du$$

$$= \frac{10}{3}\left[u\right]_0^{\pi/4}$$

$$= \frac{10}{3}\left\{\frac{\pi}{4} - 0\right\}$$

$$= \frac{10\pi}{12}.$$

12(c) $y = \frac{x^2-4}{(x-1)^2}$

$$\frac{dy}{dx} = \frac{(x-1)^2(2x) - (x^2-4)(x-1)(2)}{(x-1)^4}$$

$$= \frac{2(x-1)\left[x(x-1) - (x^2-4)\right]}{(x-1)^4}$$

$$\frac{dy}{dx} = \frac{2(4-x)}{(x-1)^3}$$

At turning points $\frac{dy}{dx} = 0$

$$\frac{2(4-x)}{(x-1)^3} = 0$$

$$x = 4.$$

When $x = 4$, $y = \frac{4^2-4}{(4-1)^2} = \frac{4}{3}$

$(4, 4/3)$ is a turning point

values of x	h	x=4	k
sign of $\frac{dy}{dx}$	+	0	−
Nature of $\frac{dy}{dx}$	/	—	\

$(4, 4/3)$ is a Maximum turning point.

As $y \to \infty$, $x = 1$

As $x \to 1$, $y = \frac{1 - 4/x^2}{1 - 2/x + 1/x^2}$

As $x \to \infty$, $y = 1$.

When $x = 0$, $y = -4$ $(0, -4)$

When $y = 0$, $\frac{x^2-4}{(x-1)^2} = 0$

$x = \pm 2$

$(-2, 0)$, $(2, 0)$

Intercept with the horizontal
when $y = 1$, $\frac{x^2-4}{(x-1)^2} = 1$

$x^2 - 4 = x^2 - 2x + 1$

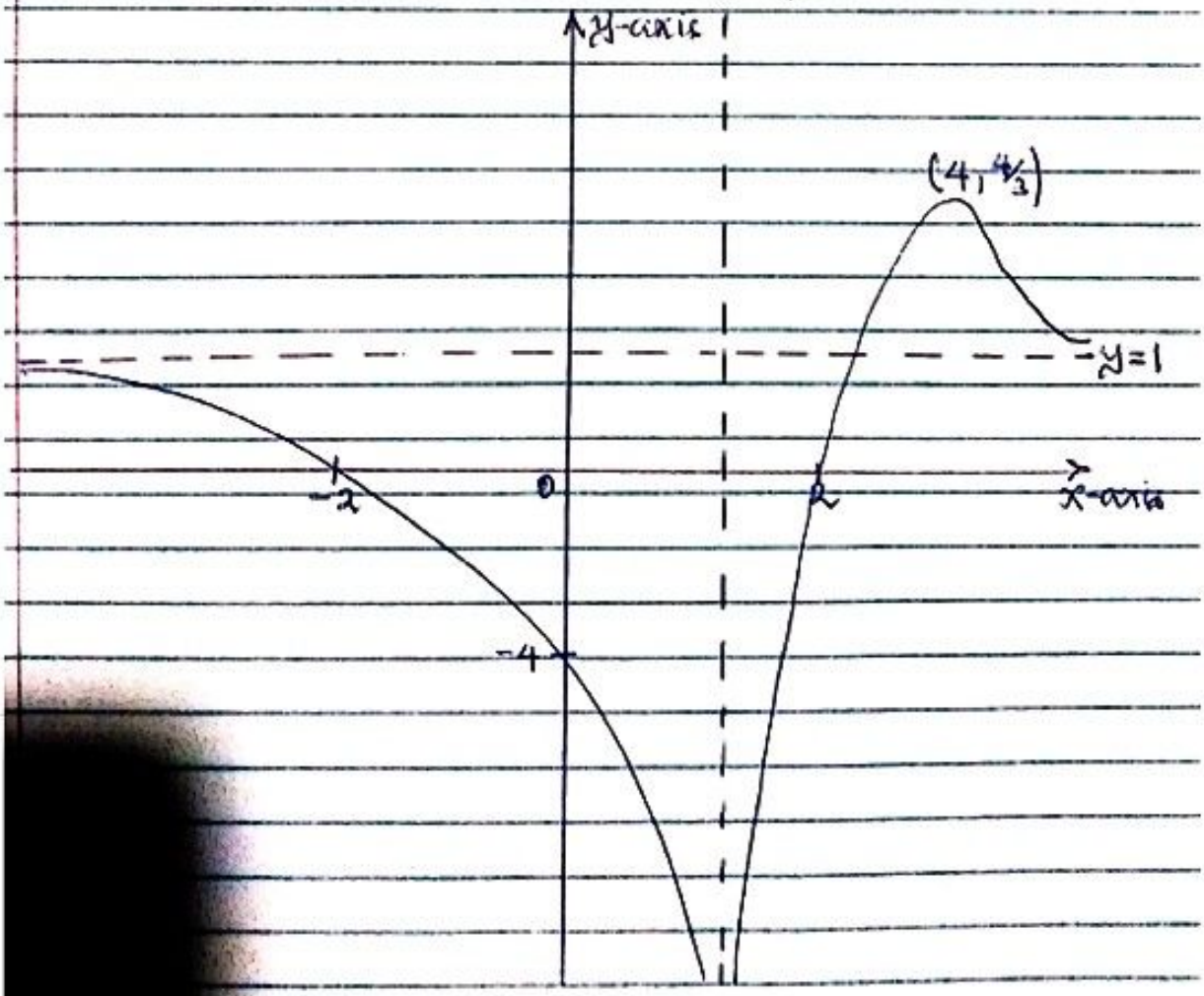
$x = 5/2$

$(5/2, 1)$

Critical values $x = -2, 1, 2$

	$x < -2$	$-2 < x < 1$	$1 < x < 2$	$x > 2$
$y = \frac{x^2-4}{(x-1)^2}$	+	−	+	+

Sketch of $y = \frac{x^2-4}{(x-1)^2}$



No. 13. (a)

$$y = x^3 + e^{3x} + 3^x$$

Let $u = x^3$, $v = e^{3x}$, $w = 3^x$

$\frac{du}{dx} = 3x^2$, $\frac{dv}{dx} = 3e^{3x}$

$\ln w = \ln 3^x$

$\ln w = x \ln 3$

$\frac{1}{w}\frac{dw}{dx} = \ln 3$

$\frac{dw}{dx} = 3^x \ln 3$

$y = u + v + w$

$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx}$

$\frac{dy}{dx} = 3x^2 + 3e^{3x} + 3^x \ln 3.$

(ii) $y = x^2 + \sin x$

$\frac{dy}{dx} = 2x + \cos x$

(iii) $y = x^3 + (\ln x)^x.$

Let $u = x^3$, $v = (\ln x)^x$

$\frac{du}{dx} = 3x^2$

$\ln v = \ln [(\ln x)^x]$

$\ln v = x \ln (\ln x)$

$\frac{1}{v}\frac{dv}{dx} = x \cdot \frac{1}{x \ln x} + \ln (\ln x)$

$\frac{dv}{dx} = \left[\frac{1}{\ln x} + \ln (\ln x)\right](\ln x)^x$

$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

$\frac{dy}{dx} = 3x^2 + \left[\frac{1}{\ln x} + \ln (\ln x)\right](\ln x)^x$

13 b) $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

$$\cos y = \frac{1-x^2}{1+x^2}$$

$$-\sin y \frac{dy}{dx} = \frac{(1+x^2)(-2x) - (1-x^2)(2x)}{(1+x^2)^2}$$

$$-\sin y \frac{dy}{dx} = \frac{2x\left[-1-x^2-1+x^2\right]}{(1+x^2)^2}$$

$$\sin y \frac{dy}{dx} = \frac{4x}{(1+x^2)^2}$$

$$\left(1-\cos^2 y\right)^{1/2} \frac{dy}{dx} = \frac{4x}{(1+x^2)^2}$$

$$\left\{1-\left(\frac{1-x^2}{1+x^2}\right)^2\right\}^{1/2} \frac{dy}{dx} = \frac{4x}{(1+x^2)^2}$$

$$\frac{2x}{1+x^2} \frac{dy}{dx} = \frac{4x}{(1+x^2)^2}$$

$$\frac{dy}{dx} = \frac{2}{1+x^2}$$

No. 15 (a)

(ii) $\int f(x)\,dx$

$$= \int 2\,dx + \int \frac{dx}{(x-1)^2} - \frac{1}{2}\int \frac{dx}{(x+1)} + \frac{5}{2}\int \frac{dx}{(x-1)}.$$

$$= 2x - \frac{1}{2}\ln(x+1) + \frac{5}{2}\ln(x-1) - \frac{1}{(x-1)} + C.$$

No. 15 (b) Let θ be the temperature of the body

$$\frac{d\theta}{dt} = -k(\theta - 20)$$

$$\int \frac{d\theta}{\theta - 20} = \int -k\,dt$$

$$\ln(\theta - 20) = -kt + C$$

$t = 0 \quad \theta = 98$

$$\ln(98 - 20) = -k(0) + C$$

$$C = \ln 78.$$

$$\ln(\theta - 20) = -kt + \ln 78$$

Initial temperature drop $= 24°/\text{min}$

$$-24 = -k(98 - 20)$$

$$k = 4/13.$$

$$\ln(\theta - 20) = -\tfrac{4}{13}t + \ln 78$$

$$\theta = 20 + 78\,e^{-\frac{4t}{13}}.$$

$\theta = ?, \quad t = 10\ \text{Min}$

$$\theta = 20 + 78\,e^{-\frac{4(10)}{13}}.$$

$\theta =$

No. 15. $f(x) = \dfrac{2x^3}{(x-1)(x^2-1)}$

$= \dfrac{2x^3}{x^3-x^2-x+1}$

$$x^3-x^2-x+1 \;\overline{)\; 2x^3} \quad \text{quotient } 2$$

$$-(2x^3-2x^2-2x+2)$$

$$2x^2+2x-2$$

$$f(x) = 2 + \frac{2x^2+2x-2}{(x-1)(x^2-1)}$$

$$= 2 + \frac{2x^2+2x-2}{(x-1)^2(x+1)}$$

Let $\dfrac{2x^2+2x-2}{(x-1)^2(x+1)} = \dfrac{A}{x+1} + \dfrac{B}{x-1} + \dfrac{C}{(x-1)^2}$.

$2x^2+2x-2 \equiv A(x-1)^2 + B(x-1)(x+1) + C(x+1)$

Let $x=1$

$2 = 2C$

$C = 1$

Let $x=-1$

$-2 = 4A$

$A = -\frac{1}{2}$

Comparing coefficients of

x^2: $2 = A + B$

$2 = -\frac{1}{2} + B$

$B = \frac{5}{2}$

$$f(x) = 2 + \frac{1}{(x-1)^2} - \frac{1}{2(x+1)} + \frac{5}{2(x-1)}$$

No. 16.

$$y = x(x-3)(x-6)$$
$$y = x^3 - 9x^2 + 18x$$
$$\frac{dy}{dx} = 3x^2 - 18x + 18$$

At turning points $\frac{dy}{dx} = 0$,

$$3x^2 - 18x + 18 = 0$$
$$x^2 - 6x + 6 = 0$$
$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(6)}}{2(1)}$$
$$x = 1.268,\ 4.732$$

When $x = 1.268$,

$$y = (1.268)^3 - 9(1.268)^2 + 18(1.268)$$
$$y = 10.392$$

$(1.268, 10.392)$

When $x = 4.732$.

$$y = (4.732)^3 - 9(4.732)^2 + 18(4.732)$$
$$y = -10.392$$

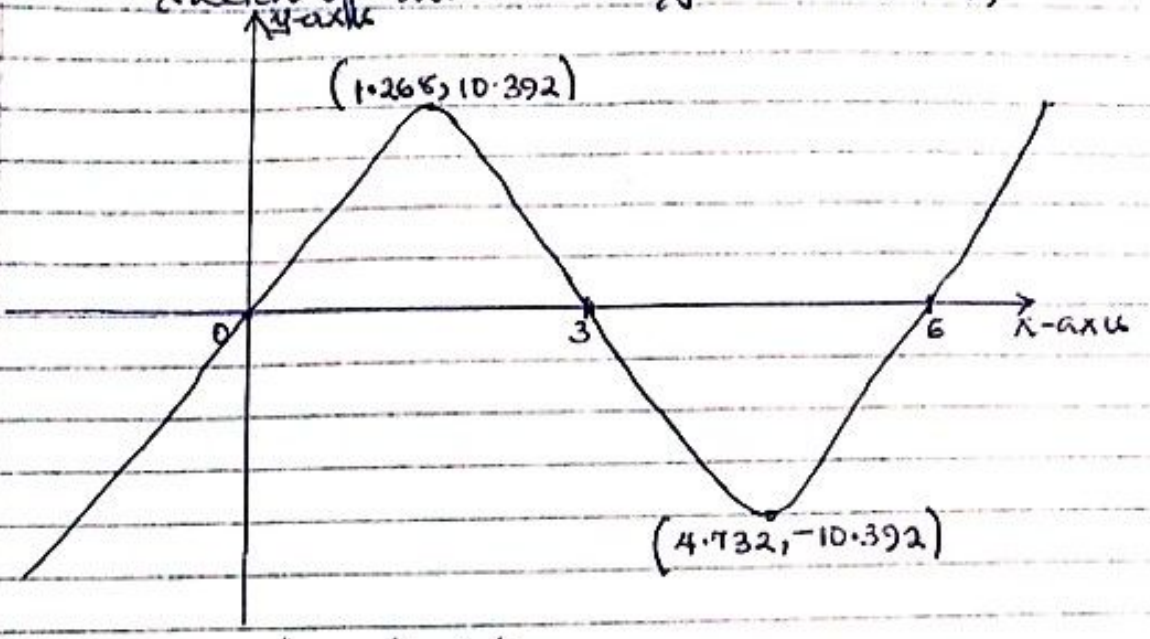
$(4.732, -10.392)$

Values of x	L	x=1.268	R	L	x=4.732	R
Sign of $\frac{dy}{dx}$	+	0	−	−	0	+
Nature of dy/dx	↗	—	↘	↘	—	↗

$(1.268, 10.392)$ is a Maximum turning point.
$(4.732, -10.392)$ is a Minimum turning point

When $x = 0$, $y = 0$ $(0,0)$
When $y = 0$, $x(x-3)(x-6) = 0$
$x = 0,\ x = 3,\ x = 6$
$(0,0)$; $(3,0)$; $(6,0)$

Sketch of the curve $y = x(x-3)(x-6)$



$A = A_1 + A_2$

$A_1 = \int_0^3 y\,dx$

$= \int_0^3 x^3 - 9x^2 + 18x\,dx$

$= \left[\frac{x^4}{4} - 3x^3 + 9x^2\right]_0^3$

$= \left(\frac{81}{4} - 81 + 81\right) - (0)$

$= \frac{81}{4}$

$A_2 = \int_3^6 y\,dx$

$= \int_3^6 x^3 - 9x^2 + 18x\,dx$

$= \left[\frac{x^4}{4} - 3x^3 + 9x^2\right]_3^6$

$= (324 - 648 + 324) - (81/4 - 81 + 81)$

$= \frac{-81}{4}$

$A = \frac{81}{4} + \frac{81}{4}$

$= \frac{81}{2}$ Sq units.

16 6) (i) $x^2 \frac{dy}{dx} = x^2 + y^2$.

$\frac{dy}{dx} = 1 + \left(\frac{y}{x}\right)^2$.

$y = vx \Rightarrow v = y/x$.

$\frac{dy}{dx} = v + x\frac{dv}{dx}$.

$v + x\frac{dv}{dx} = 1 + v$

$x\frac{dv}{dx} = 1$

$\int dv = \int \frac{dx}{x}$

$v = \ln kx$.

$\frac{y}{x} = \ln kx$.

$y = x \ln kx$.

(ii) Let x be the distance yet to be covered.

$\frac{dx}{dt} = -k\sqrt{x}$.

$\int \frac{dx}{\sqrt{x}} = \int -k\,dt$

$2\sqrt{x} = -kt + c$.

$t = 0$, $x = 4000$ m.

$2\sqrt{400} = -k(0) + c$

$c = 40$.

$2\sqrt{x} = -kt + 40$.

$t = 0 \quad \frac{dx}{dt} = -16 \quad x = 400$.

$-16 = -k\sqrt{400}$

$k = 4/5$

$$2\sqrt{x} = \frac{-4}{5}t + 40.$$

When the distance yet to be covered is zero Opio has reached school

$$2\sqrt{0} = \frac{-4}{5}t + 40.$$

$$t = 50$$

No. 17 (a)

For $y = x^2 - 4x + 3$

$\frac{dy}{dx} = 2x - 4$

At turning point $\frac{dy}{dx} = 0$

$2x - 4 = 0$

$x = 2$

When $x = 2$, $y = (2)^2 - 4(2) + 3 = -1$. $(2, -1)$

$\frac{d^2y}{dx^2} = 2 > 0$

$(2, -1)$ is a Minimum turning point

When $x = 0$, $y = B$ $(0, 3)$

When $y = 0$, $x^2 - 4x + 3 = 0$

$(x - 3)(x - 1) = 0$

$x = 1, x = 3$

$(1, 0), (3, 0)$

For $y = 9 - x^2$

$\frac{dy}{dx} = -2x$

At turning point $\frac{dy}{dx} = 0$

$-2x = 0$

$x = 0$

When $x = 0$, $y = 9$ $(0, 9)$

$\frac{d^2y}{dx^2} = -2 < 0$

$(0, 9)$ is a Maximum turning point

When $x = 0$, $y = 9$ $(0, 9)$

When $y = 0$, $9 - x^2 = 0$

$x = \pm 3$

$(-3, 0), (3, 0)$.

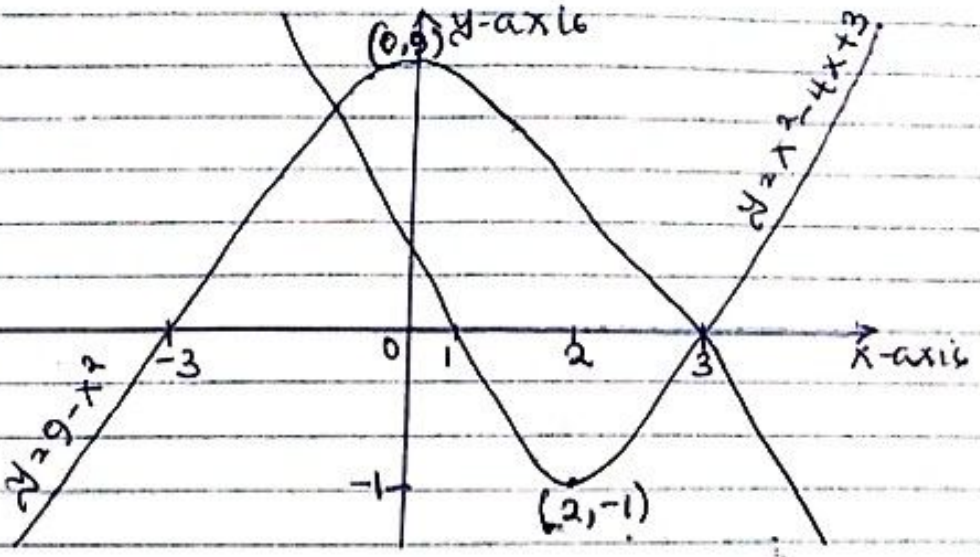
Point of Intersection

$x^2 - 4x + 3 = 9 - x^2$

$(x - 3)(x + 1) = 0$

$x = -1, x = 3$.

Sketch of $y = x^2 - 4x + 3$ and $y = 9 - x^2$



17 a)(ii) $A = \int_{-1}^{3} (9 - x^2) - (x^2 - 4x + 3)\, dx$

$= \int_{-1}^{3} 6 + 4x - 2x^2 dx$

$= \left[6x + 2x^2 - \frac{2x^3}{3} \right]_{-1}^{3}$

$= \left(18 + 18 - 18\right) - \left(-6 + 2 + \frac{2}{3}\right)$

$= \frac{64}{3}$ Sq Units.

17 a iii) $V = \pi \int_{-1}^{3} (9 - x^2)^2 - (x^2 - 4x + 3)^2\, dx.$

$= \pi \int_{-1}^{3} 8x^3 - 43x^2 + 24x + 72\, dx$

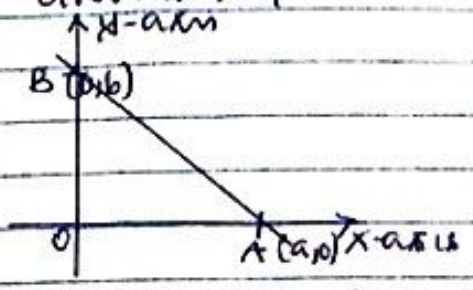
$= \pi \left[2x^4 - \frac{43}{3}x^3 + 12x^2 + 72x \right]_{-1}^{3}$

$= \pi \left\{ \left(162 - 387 + 108 + 216\right) - \left(2 + \frac{43}{3} + 1\right) \right.$

$= \frac{856\pi}{6}$ Cubic Units

GEOMETRY

No. 18



Gradient $AB = \frac{b-0}{0-a} = \frac{-b}{a} = -\frac{4}{3}$

$a = \frac{3b}{4}$

Length $AB = \sqrt{(a-0)^2 + (0-b)^2} = 10$

$a^2 + b^2 = 100$

$\left(\frac{3b}{4}\right)^2 + b^2 = 100$

$\frac{25b^2}{4} = 100$

$b^2 = 16$

$b = \pm 4$ Discard $b = -4$

$b = 4$

$a = 3$

Equation of line

$y = -\frac{4}{3}x + 4$

b) $x = r\cos\theta$, $y = r\sin\theta$

$x^2 + y^2 = 9$

$r^2\cos^2\theta + r^2\sin^2\theta = 9$

$r^2(\cos^2\theta + \sin^2\theta) = 9$

$r^2 = 9$

$r = 3$.

18 b(ii) $y^2 = 4x$

$(r\sin\theta)^2 = 4r\cos\theta$

$r = 4\cot\theta\operatorname{cosec}\theta$.

18 (c) $y^2 = 8x + 16$

$(y-0)^2 = 8(x+2)$.

Vertex $(-2,0)$

$4a = 8$

$a = 2$

Focus $= (0,0)$

Directrix, $x = -4$.

$y^2 - 8y = 4x$.

$(y-4)^2 = 4x + 16$

$(y-4)^2 = 4(x+4)$

Vertex $= (-4,4)$

$4a = 4$

$a = 1$

Focus $= (-3,4)$

Directrix, $x = -5$

18 (d).

$A(2,0)$ $P(x,y)$ $M(-2,y)$

$\sqrt{(x-2)^2 + (y-0)^2} = \sqrt{(x+2)^2 + (y-y)^2}$.

$x^2 - 4x + 4 + y^2 = x^2 + 4x + 4$

$y^2 = 8x$.

ii) $\sqrt{x^2+y^2} - 1 = 3 - x$.

$\sqrt{x^2+y^2} = 4 - x$.

$x^2 + y^2 = 16 - 8x + x^2$

$y^2 = -8(x-2)$.

No. 19

A(-3,1) B(2,3) and C(4,-2).

For A(-3,1) $x^2+y^2+2gx+2fy+c=0$

$(-3)^2+(1)^2+2g(-3)+2f(1)+c=0$

$-6g+2f+c=-10$ ---(i)

For B(2,3) $(2)^2+(3)^2+2g(2)+2f(3)+c=0$

$4g+6f+c=-13$ ---(ii)

For C(4,-2) $4^2+(-2)^2+2g(4)+2f(-2)+c=0$

$8g+-4f+c=-20$ ---(iii)

Eq(i) - Eq(ii) $\Rightarrow$ $10g+4f=-3$ --(iv)

Eq(ii) - Eq(iii) $\Rightarrow$ $4g-10f=-7$ --(v)

$2\times$Eq(iv) $-5\times$Eq(v) $\Rightarrow$ $58f=29$

$f=1/2$.

$4g-10(1/2)=-7$

$g=-1/2$.

$8(-1/2)-4(1/2)+c=-20$

$c=-14$

Equation of the circle is

$x^2+y^2-x+y-14=0$.

19(b) $x^2+y^2-6x+8=0$

$(x-3)^2+(y-0)^2=1$

Centre (3,0)

radius = 1.

$x^2+y^2-4x+2y+6=0$

$(x-2)^2+(y+1)^2=6+5$.

Centre (2,-1)

radius $=\sqrt{6+5}$

distance $=\sqrt{(3-2)^2+(0-(-1))^2}$

$=\sqrt{2}$.

For orthogonal circles

$$r_1^2 + r_2^2 = d^2.$$

$$1^2 + (\sqrt{b+5})^2 = (\sqrt{2})^2.$$

$$b = -4.$$

Radius of the circle C_2

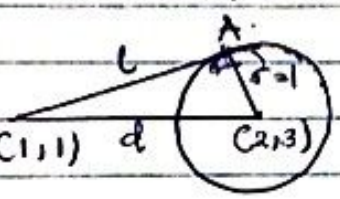
$$r = \sqrt{5-4}$$

$$r = 1 \text{ Units}$$

19 (c) $x^2 + y^2 - 4x - 6y + 12 = 0$

$$(x-2)^2 + (y-3)^2 = 1$$

Centre $\cdot C(2,3)$, Radius $= 1$.



$$d = \sqrt{(2-1)^2 + (3-1)^2}$$

$$d = \sqrt{5}$$

$$b = \sqrt{d^2 - r^2}$$

$$b = \sqrt{(\sqrt{5})^2 - 1}$$

$$b = 2 \text{ Units}.$$

(ii) Let $y = mx + c$ be the tangent to the circle

$$1 = \left| \frac{2m + c - 3}{\sqrt{m^2+1}} \right|$$

$$m^2 + 1 = (2m + c - 3)^2 \quad \text{---} (i)$$

$P(1,1)$, $1 = m + c$, $c = 1 - m$ --- (ii)

Solving Eq (i) & Eq (ii) Simultaneously

$$m^2 + 1 = (m-2)^2$$

$$m = 3/4$$

$$c = 1/4$$

$$y = \frac{3}{4}x + \frac{1}{4}$$

No. 20

$y^2 = 4ax$

$2y \frac{dy}{dx} = 4a$

$\frac{dy}{dx} = \frac{2a}{y}$

$\left.\frac{dy}{dx}\right|_{(ap^2, 2ap)} = \frac{2a}{2ap} = \frac{1}{p}$

Equation of OQ is $y = \frac{1}{p}x$.

Point of intersection of OQ and $y^2 = 4ax$

$y = \frac{1}{p}x, \quad y^2 = 4ax$

$\frac{x^2}{p^2} = 4ax$

$x^2 - 4ap^2 x = 0$

$x(x - 4ap^2) = 0$

$x = 0, \quad x = 4ap^2$

When $x = 0$, $y = 0$ $O(0,0)$.

$y = \frac{1}{p}(4ap^2)$

$y = 4ap$

$Q(4ap^2, 4ap)$

Mid point $M = (2ap^2, 2ap)$

Gradient $PM = \frac{2ap - 2ap}{2ap^2 - ap^2} = 0$

Since Gradient PM is 0, it is perpendicular to the y-axis.

ALGEBRA

No 25 (a)

$$2x - y + z = 6 \quad \cdots (i)$$
$$3x + y - z = 4 \quad \cdots (ii)$$
$$4x + y - 2z = 3 \quad \cdots (iii)$$

Eq (i) + Eq (ii) $\Rightarrow$ $5x = 10$

$$x = 2$$

Eq (i) + Eq (iii) $\Rightarrow$ $6x - z = 9$

$$6(2) - z = 9$$
$$z = 3$$
$$2(2) - y + 3 = 6$$
$$y = 1.$$

(b) $x^2 + 2x + 8 = 0$

$$\alpha + \beta = -2$$
$$\alpha\beta = 8$$
$$\alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$$
$$= (-2)^3 - 3(8)(-2)$$
$$= 40.$$

ii) $$\frac{\alpha}{\beta^2+1} + \frac{\beta}{\alpha^2+1}$$

$$\Rightarrow \frac{\alpha^3 + \beta^3 + \alpha + \beta}{\alpha^2 + \beta^2 + (\alpha\beta)^2 + 1}$$

$$\Rightarrow \frac{(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) + (\alpha+\beta)}{(\alpha+\beta)^2 - 2\alpha\beta + (\alpha\beta)^2 + 1}$$

$$\Rightarrow \frac{40 + (-2)}{(-2)^2 - 2(8) + 8^2 + 1}$$

$$= \frac{38}{53}$$

$$\left(\frac{\alpha}{\beta^2+1}\right)\left(\frac{\beta}{\alpha^2+1}\right)$$

$$= \frac{\alpha\beta}{(\alpha+\beta)^2 - 2\alpha\beta + (\alpha\beta)^2 + 1}$$

$$= \frac{-2}{(53)}$$

Equation whose roots are α/β^2+1 and β/α^2+1 is $x^2 - \frac{38}{53}x - \frac{2}{53} = 0$.

c) $S_n = n(n+\alpha+1)$

$U_n = S_n - S_{n-1}$

$= n(n+\alpha+1) - (n-1)(n+\alpha)$

$U_n = 2n+\alpha$.

$d = U_n - U_{n-1}$

$= (2n+\alpha) - (2(n-1)+\alpha)$

$= 2$.

Since the common difference exists between two consecutive terms, the Progression is Arithmetic.

ii) $U_6 = 2(6) + \alpha = 14$

$\alpha = 2$.

$S_n = n(n+3)$

$S_{20} = 20(23)$

$S_{20} = 460$.

$U_n = 2n+2$

$U_{20} = 2(20)+2$

$U_{20} = 42$.

No. 25

(d)

Year of Deposit	Amount that accumulate at the end of the 12th yr
12th	$1000\left(1+4/100\right)^{1}$
11th	$1000\left(1+4/100\right)^{2}$
10th	$1000\left(1+4/100\right)^{3}$
--	-- -- --
2nd	$1000\left(1+4/100\right)^{11}$
1st	$1000\left(1+4/100\right)^{12}$

Total Amount $= 1000\times1.04 + 1000(1.04)^2 + \cdots + 1000(1.04) + 1000(1.04)^{12}$

$$= 1000\times1.04\left[\frac{1-(1.04)^{12}}{1-(1.04)}\right]$$

$$= 15,626.838$$

No. 26

$$\frac{x+y}{2} = \frac{y+z}{3} = \frac{x+z}{4} = t$$

$$\frac{(x+y)+(y+z)+(x+z)}{2+3+4} = t$$

$$\frac{2x+2y+2z}{9} = t$$

$$x+y+z = 6$$

$$9t = 12$$

$$t = 4/3$$

$$x+y = 8/3 \quad \text{---(i)}$$

$$y+z = 4 \quad \text{---(ii)}$$

Eq(i) − Eq(ii) $\quad x-z = -4/3 \quad \text{---(iv)}$

$$x+z = 16/3 \quad \text{---(iii)}$$

+ Eq(iv) $\Rightarrow \quad 2x = 4$

$$x = 2$$

$$z = 2/3$$

$$y = 10/3$$

No. 26

b) $\frac{x-\lambda}{1-\lambda} = \frac{1}{1-\lambda} + \frac{2}{x}$

$\frac{x-\lambda-1}{1-\lambda} = \frac{2}{x}$

$x^2 - (\lambda+1)x + (2\lambda-2) = 0$

For repeated roots $b^2-4ac = 0$

$[-(1+\lambda)]^2 - 4(1)(2\lambda-2) = 0$

$(\lambda-3)^2 = 0$

$\lambda = 3$.

c) $P(x) = x^4 + ax + b$

$(x-1)^2 = 0$

$x = 1$

$P(1) = 1 + a + b = 0$

$a + b = -1$ --①

$P'(x) = 4x^3 + a$

$P'(1) = 4 + a = 0$

$a = -4$

$b = 3$

$P(x) = x^4 - 4x + 3$

$P(1) = 0$.

No. 28.

$${}^{n}P_6 = 48\ {}^{n}C_4$$

$$\frac{n!}{(n-6)!} = 48\,\frac{n!}{(n-4)!\,4!}$$

$$\frac{(n-4)!}{(n-6)!} = \frac{48}{24}$$

$$(n-4)(n-5) = 2$$

$$n^2 - 9n + 18 = 0$$

$$(n-6)(n-3) = 0$$

$n = 6$ or $n = 3$

Discard $n = 3$

$n = 6$.

b) R-2, E = 3, F=1, N-1, D-1, U-1, M=1

No of arrangements of RREFERENDUM $= \frac{10!}{2!\,3!}$

$= 302,400$

ii) No of arrangements when two letters D & M are together $= \frac{9!}{2!\,3!} \times 2$

$= 60,480$

(c)

Sample space	4 men	5 women
Selection	3 men	3 women
No of choices	${}^4C_3$	${}^5C_3$

Total no of choices $= {}^4C_3 \times {}^5C_3$

$= 4 \times 10$

$= 40$

ii) More men than women.

Sample space	4 men	5 women
Selection	4 men	2 women
No of choices	$^4C_4$	$^5C_2$

Number of choices $= {}^4C_4 \times {}^5C_2$

$= 1 \times 10$

$= 10$.

30 b(ii) $\frac{yi}{x+4i} = \frac{2+i}{3-2i}$

$3yi + 2y = (2+i)(x+4i)$

$2y + 3yi = 2x - 4 + (x+8)i$

$2y = 2x - 4$

$y = x - 2 \quad --(1)$

$3y = (x+8)$

$3(x-2) = x+8$

$x = 7$

$y = 5$

c) $\frac{z + 2i}{z - 3} = \lambda$

$\frac{x + yi + 2i}{x + yi - 3} = \lambda$

$x + (y+2)i = (x-3)\lambda + \lambda yi$

$\frac{x}{x-3} = \frac{y+2}{y}$

$xy = xy + 2x - 3y - 6$

$2x - 3y = 6$

No. 30.

$$z = \frac{(3+2i)(2+5i)}{2-i}$$

$$= \frac{-6+14i}{2-i}$$

$$= \frac{(-6+14i)(2+i)}{(2-i)(2+i)}$$

$$= \frac{-12+28i-6i-14}{2^2+1^2}$$

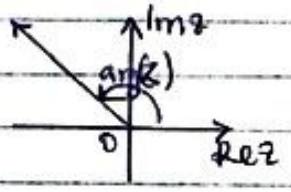
$$z = -\frac{26}{5} + \frac{22}{5}i$$

$$|z| = \sqrt{\left(\frac{-26}{5}\right)^2 + \left(\frac{22}{5}\right)^2}$$

$$= \sqrt{46.4}$$

$$= 6.81175$$

$$\arg(z) = \tan^{-1}\left(\frac{\frac{22}{5}}{\frac{-26}{5}}\right)$$



$$= 180^\circ - \tan^{-1}\left(\frac{11}{13}\right)$$

$$= 180^\circ - 40.24^\circ$$

$$= 139.76^\circ$$

$$z = 6.81175\left(\cos 139.76^\circ + i \sin 139.76^\circ\right)$$

(b) $z = 3+2i$ is a root,

$3-2i$ is also a root.

Eqn. with roots $3+2i$ & $3-2i$

$$z^2 - 6z + 13 = 0$$

$$z^3 - 5z^2 + 7z + 13 = (z+1)(z^2-6z+13) = 0$$

roots of eqn. are $z = -1, 3 \pm 2i$

31 (c) $z = \cos\theta + i\sin\theta$

$z^n = (\cos\theta + i\sin\theta)^n$

$= \cos n\theta + i\sin n\theta$

$\frac{1}{z^n} = z^{-n}$

$= (\cos\theta + i\sin\theta)^{-n}$

$= (\cos n\theta - i\sin n\theta)$

$z^n + 1/z^n = \cos n\theta + i\sin n\theta - (\cos n\theta - i\sin n\theta)$

$z^n - 1/z^n = 2i\sin n\theta$

For $z^n - 1/z^n = 2i\sin n\theta$

$n=1 \quad z - 1/z = 2i\sin\theta$

$n=3 \quad z^3 - 1/z^3 = 2i\sin 3\theta$

$\left(z - \frac{1}{z}\right)^3 = (2i\sin\theta)^3$

$8i\sin^3\theta = z^3 + 3z^2(-1/z) + 3z(-1/z)^2 + (-1/z)^3$

$= z^3 - 3z + 3/z - 1/z^3$

$= \left(z^3 - 1/z^3\right) - 3\left(z - 1/z\right)$

$8i\sin^3\theta = 2i\sin 3\theta - 3(2i\sin\theta)$

$8\sin^3\theta = 2\sin 3\theta - 6\sin\theta$

$\sin^3\theta = \frac{1}{4}(\sin 3\theta - 3\sin\theta)$

No. 31.

$$z_1 = (5+2i)(6+i)$$
$$= 28+17i$$
$$z_2 = (5-2i)(6-i)$$
$$28-17i$$

$$28^2+17^2 = (28+17i)(28-17i)$$
$$= (5+2i)(6+i)(5-2i)(6-i)$$
$$= (5^2+2^2)(6^2+1^2)$$
$$= 29 \times 37.$$

8 (b) $z\bar{z} + 2z = 9+2i$

$z = x+yi$

$(x+yi)(x-yi) + 2(x+yi) = 9+2i$

$x^2+y^2+2x+2yi = 9+2i$

$2y = 2$

$y = 1$

$x^2+y^2+2x = 9$

$x^2+(1)^2+2x = 9$

$x^2+2x-8 = 0$

$(x+4)(x-2) = 0$

$x = 2, \; x = -4.$

$z = 2+i$ or $z = -4+i$

(ii) $$\frac{\cos\frac{5\pi}{17} + i\sin\frac{5\pi}{17}}{\left(\cos\frac{4\pi}{17} - i\sin\frac{4\pi}{17}\right)^3}$$

$$\frac{(\cos\pi + i\sin\pi)^{5/17}}{\left((\cos\pi + i\sin\pi)^{-4/17}\right)^3}$$

$$\frac{(\cos\pi + i\sin\pi)^{5/17 + 12/17}}{(\cos\pi + i\sin\pi)^1}$$

$$-1+0i$$

Vertical Asymptote

As $y \to \infty$ $x = 3$

Oblique Asymptote

$$y = \frac{x(x-3) + 3(x-3) + 9}{x-3}$$

$$y = (x+3) + \frac{9}{x-3}$$

As $x-3 \to \infty$, $y = x+3$

Critical values of $x = 0, 3$

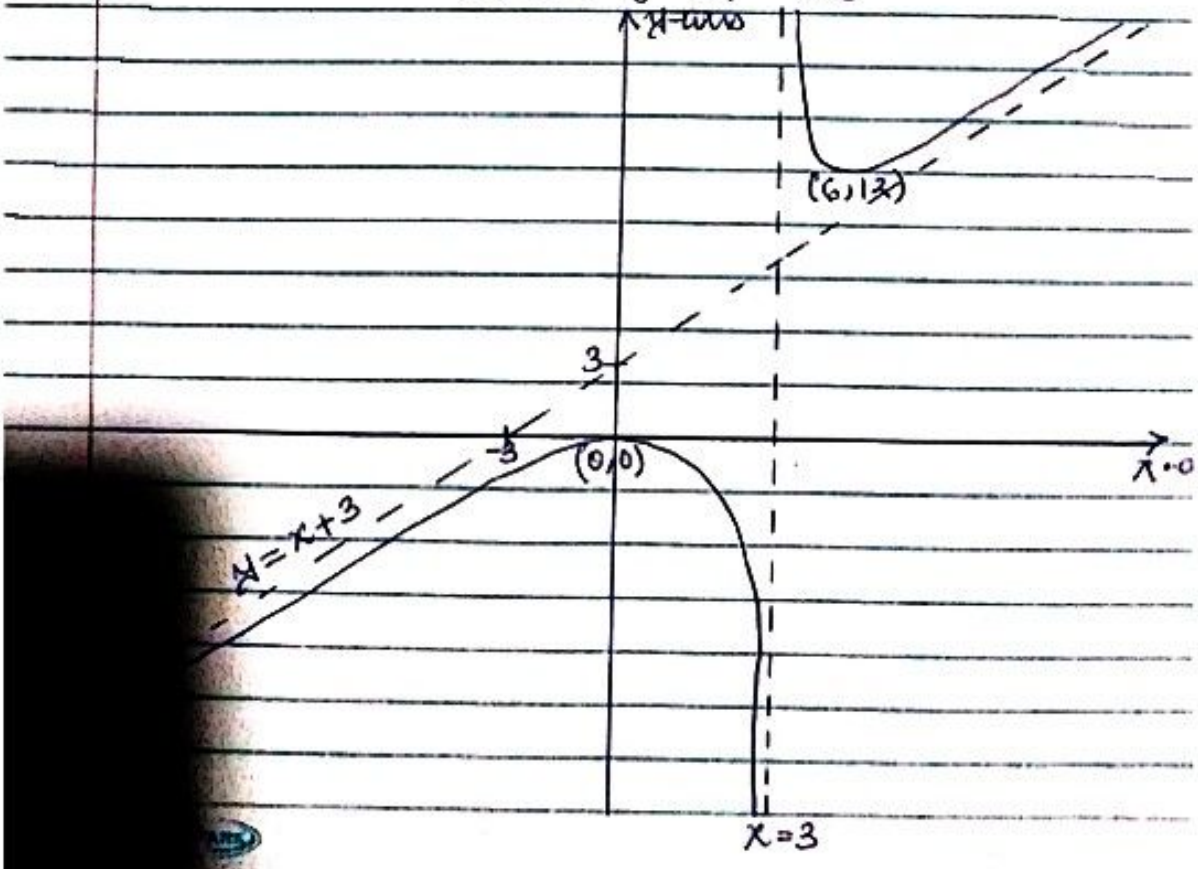
	$x<0$	$0<x<3$	$x>3$
$y = \frac{x^2}{x-3}$	−	−	+

For $y = x+3$

When $x = 0$, $y = 3$ $(0,3)$

When $y = 0$, $x = -3$ $(-3,0)$

Sketch of $y = \frac{x^2}{x-3}$.



VECTORS

34a) $\overrightarrow{OP} = 4i + 3j + 2k$, $\overrightarrow{OR} = 5i - 2j + 2k$

$\overrightarrow{OQ} = 3i + j + 4k$

$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$

$= \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}$

$\overrightarrow{QR} = \overrightarrow{OR} - \overrightarrow{OQ}$

$= \begin{pmatrix} 5 \\ -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \\ -2 \end{pmatrix}$

$\overrightarrow{PQ} \cdot \overrightarrow{QR} = \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -3 \\ -2 \end{pmatrix}$

$= -2 + 6 - 4$

$= 0$

Since $\overrightarrow{PQ} \cdot \overrightarrow{QR} = 0$, $\angle PQR = 90^\circ$

$\overrightarrow{PQ} = \overrightarrow{SR} = \overrightarrow{OR} - \overrightarrow{OS}$

$\begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \\ 2 \end{pmatrix} - \overrightarrow{OS}$

$\overrightarrow{OS} = \begin{pmatrix} 5 \\ -2 \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}$

$\overrightarrow{OS} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$

$S = (6, 0, 0)$

34b) $\begin{pmatrix} 1 + 2\lambda \\ 2 + \lambda \\ p - \lambda \end{pmatrix} = \begin{pmatrix} 2 + \mu \\ -2 + 2\mu \\ 4 - \mu \end{pmatrix}$

$2\lambda - \mu = 1$

$\lambda - 2\mu = -4$

$\lambda = 2$

$\mu = 3$

$p - 2 = 4 - 3$

$p = 3$

$$\overrightarrow{OQ} = \begin{pmatrix} 2+(3) \\ -2+2(3) \\ 4-(3) \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 1 \end{pmatrix}$$

$$Q = (5, 4, 1)$$

34 6(iii) $\quad \underset{\sim}{d_1} = 2\underset{\sim}{i} + \underset{\sim}{j} - \underset{\sim}{k}$

$$\underset{\sim}{d_2} = \underset{\sim}{i} + 2\underset{\sim}{j} - \underset{\sim}{k}$$

$$\underset{\sim}{d_1} \cdot \underset{\sim}{d_2} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$= 2 + 2 + 1$$

$$= 5$$

$$|\underset{\sim}{d_1}| = \sqrt{2^2 + 1^2 + (-1)^2} = \sqrt{6}$$

$$|\underset{\sim}{d_2}| = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{6}$$

$$\cos\theta = \frac{\underset{\sim}{d_1} \cdot \underset{\sim}{d_2}}{|\underset{\sim}{d_1}||\underset{\sim}{d_2}|}$$

$$\cos\theta = \frac{5}{\sqrt{6}\sqrt{6}}$$

$$\theta = \cos^{-1}(5/6)$$

$$\theta = 33.56^\circ.$$

36 (i) $\underset{\sim}{a} = 2\underset{\sim}{i} + 3\underset{\sim}{j} + 2\underset{\sim}{k}$

$\underset{\sim}{b} = \underset{\sim}{i} + 3\underset{\sim}{j} + \underset{\sim}{k}$

$\underset{\sim}{c} = \underset{\sim}{i} - 3\underset{\sim}{j} + \underset{\sim}{k}$

$$|\underset{\sim}{a} + 2\underset{\sim}{b} + 3\underset{\sim}{c}| = \left|\begin{pmatrix}2\\3\\2\end{pmatrix} + 2\begin{pmatrix}1\\3\\1\end{pmatrix} + 3\begin{pmatrix}1\\-3\\1\end{pmatrix}\right|$$

$$= \left|\begin{pmatrix}7\\0\\7\end{pmatrix}\right|$$

$$= \sqrt{7^2 + 0^2 + 7^2}$$

$$= 7\sqrt{2}$$

a(ii)

$$\underset{\sim}{a} \times \underset{\sim}{b} = \begin{vmatrix}\underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k}\\2 & 3 & 2\\1 & -3 & 1\end{vmatrix}$$

$$= 9\underset{\sim}{i} - 9\underset{\sim}{k}$$

$$(\underset{\sim}{a} \times \underset{\sim}{b}) \cdot \underset{\sim}{c} = \begin{pmatrix}9\\0\\-9\end{pmatrix} \cdot \begin{pmatrix}1\\-3\\1\end{pmatrix}$$

$$= 9 + 0 - 9$$

$$(\underset{\sim}{a} \times \underset{\sim}{b}) \cdot \underset{\sim}{c} = 0$$

Since $(a \times b) \cdot c = 0$, The vectors $\underset{\sim}{a}, \underset{\sim}{b}$ and $\underset{\sim}{c}$ are coplanar.

a(iii) $\underset{\sim}{r} \cdot \underset{\sim}{n} = \underset{\sim}{a} \cdot \underset{\sim}{n}$

$$\begin{pmatrix}x\\y\\z\end{pmatrix} \cdot \begin{pmatrix}9\\0\\-9\end{pmatrix} = \begin{pmatrix}3\\2\\-1\end{pmatrix} \cdot \begin{pmatrix}9\\0\\-9\end{pmatrix}$$

$$9x - 9z = 27 + 18$$

$$9x - 9z = 45$$

$$x - z = 5.$$

b) $3x + y + z = 7$ ---(i)

$x + y - z = 3$ ---(ii)

Eq(i) + Eq(ii) $\Rightarrow$ $4x + 2y = 10$

$y = 5 - 2x$

Set $x = t$

$y = 5 - 2t$

$x + y - z = 3$

$t + 5 - 2t - z = 3$

$z = 2 - t$

$$t = \frac{x}{1} = \frac{y-5}{-2} = \frac{z-2}{-1}$$

$$x = \frac{y-5}{-2} = \frac{z-2}{-1}$$

35 b(ii)

$$\underset{\sim}{n_1} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \qquad \underset{\sim}{n_2} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$\underset{\sim}{n_1} \cdot \underset{\sim}{n_2} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$= 3 + 1 - 1$$

$$= 3$$

$$|\underset{\sim}{n_1}| = \sqrt{3^2 + 1^2 + 1^2} = \sqrt{11}$$

$$|\underset{\sim}{n_2}| = \sqrt{1^2 + 1^2 + (-1)^2} = \sqrt{3}$$

$$\cos\theta = \frac{\underset{\sim}{n_1} \cdot \underset{\sim}{n_2}}{|\underset{\sim}{n_1}||\underset{\sim}{n_2}|}$$

$$\cos\theta = \frac{3}{\sqrt{3}\sqrt{11}}$$

$$\theta = \cos\left(\frac{3}{\sqrt{33}}\right)$$

$$\theta = 58.52^\circ$$

Acute angle between the planes is 58.52°.